Do bridge-monitoring error bars survive an unseen season? Physics-guided, conformally calibrated temperature compensation on the KW51 railway bridge

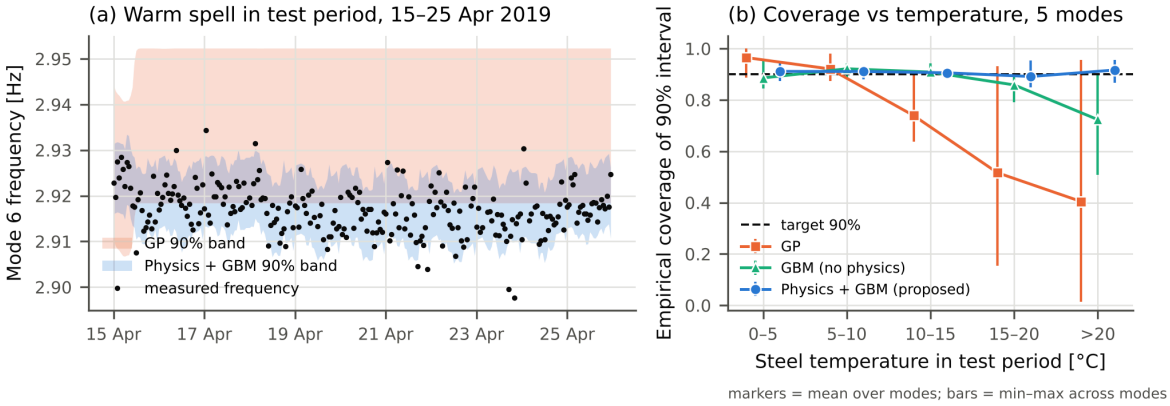
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PROBLEM

Structural health monitoring tracks a bridge's natural frequencies, because damage lowers stiffness and therefore frequency. Temperature changes frequency by a similar amount, so the measurement must be *compensated*, and the alarm threshold is only as trustworthy as its uncertainty band. Conformal prediction gives bands with a coverage guarantee, but only if past and future data are exchangeable. Seasonal data are not. **How much does coverage degrade when a model must predict a season it has not seen, and does a physics prior help?**

DATA AND METHOD

I used the public KW51 dataset (Maes & Lombaert, Zenodo): hourly natural frequencies of a steel railway bridge in Leuven, Belgium, Oct 2018–Jan 2020, retrofitted in May–Sep 2019. No new experiments were run. I trained on Nov–Jan, calibrated on Feb, and tested on Mar–mid-May, when temperatures reached 26 °C against a training maximum of 13.6 °C. The proposed model is a 4-parameter thermo-elastic curve, $f = f_0\sqrt{(1+a\Delta T)}$ plus a cold-stiffening term, with a small gradient-boosted model fitted to its residuals. It was compared with linear, cubic, Gaussian-process and physics-free boosted-tree baselines, all wrapped in the same split-conformal procedure. Five vibration modes were analysed.



Key figure. (a) Mode 6 during a warm spell in the test period: the Gaussian-process 90 % band drifts away from the measurements, while the physics-guided band follows them. (b) Empirical coverage of nominal 90 % intervals by test temperature, averaged over 5 modes (bars: min–max across modes). Training never exceeded 13.6 °C.

RESULTS SO FAR

- With a random (exchangeable) split, every model reached 90 % coverage, as theory predicts.
- With the seasonal split, mean coverage was 0.91 for the physics-guided model, 0.90 for physics-free boosted trees, 0.83 for the cubic and 0.79 for the Gaussian process. Above 20 °C, the physics-guided model held 0.87–0.96 across modes, while the GP fell to 0.01–0.07 on three modes and boosted trees to 0.51–0.79 on four.
- The fitted temperature coefficient was negative for 4 of 5 modes, consistent with steel softening as it warms. The cold-stiffening threshold was about 1 °C for all modes.
- Frozen conformal bands flagged the 2019 retrofit on 100 % of post-retrofit days for 3 modes, with 0–1.3 % false-alarm days for the physics-guided model.

LIMITATIONS AND NEXT STEPS

The extrapolation result rests on only 8 days above 20 °C and one bridge. The retrofit was a large stiffening, not damage, and it does not separate the methods by detection power. No test data fell below 0 °C. Adaptive conformal inference restores coverage for every model, but it would also adapt away real damage. Next I will run a synthetic minimum-detectable-shift study and rolling-origin evaluation, and add a physics-informed neural-network residual model. **I would value feedback on the physical interpretation of the cold-stiffening term and on evaluation design.**

All numbers are reproducible from the project repository. Prior work on KW51 (Maes et al. 2022; Wittenberg et al. 2025; Neumann et al. 2025–26) covers environmental compensation and monitoring. In my search I did not find conformal coverage evaluated under seasonal shift for this dataset.